

Kubernetes Practice

Replicaset:

A **ReplicaSet** in Kubernetes is used to **maintain a stable set of replica Pods running at any given time**

If a Pod crashes → ReplicaSet creates a new Pod

If a Pod is deleted → ReplicaSet creates a new Pod

If you want 5 Pods → It ensures exactly 5 Pods are running

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: ipl-rs
  namespace: dharani-ns
spec:
  replicas: 5
  selector:
    matchLabels:
      app: ipl
  template:
    metadata:
      labels:
        app: ipl
    spec:
      containers:
        - name: ipl-container
          image: muralisocial123/ipl:latest
          ports:
            - containerPort: 3000
```

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```
controlplane:~/dharani-k8s$ ls
controlplane:~/dharani-k8s$ vi rs.yaml
controlplane:~/dharani-k8s$ ls
rs.yaml
controlplane:~/dharani-k8s$ cat re.yaml
cat: re.yaml: No such file or directory
controlplane:~/dharani-k8s$ cat rs.yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: ipl-rs
  namespace: dharani-ns
spec:
  replicas: 5
  selector:
    matchLabels:
      app: ipl
  template:
    metadata:
      labels:
        app: ipl
    spec:
      containers:
        - name: ipl-container
          image: muralisocial123/ipl:latest
          ports:
            - containerPort: 3000
controlplane:~/dharani-k8s$
```

rs.yaml

controlplane:~/dharani-k8s\$ kubectl get pod

NAME	READY	STATUS	RESTARTS
ipl-rs-dzdcz	1/1	Running	0
ipl-rs-hbwtb	1/1	Running	0
ipl-rs-jg9gv	1/1	Running	0
ipl-rs-pvxcn	1/1	Running	0
ipl-rs-qtgt8	1/1	Running	0

controlplane:~/dharani-k8s\$ kubectl get rs

NAME	DESIRED	CURRENT	READY	AGE
ipl-rs	5	5	5	2m1s

controlplane:~/dharani-k8s\$ kubectl get all

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rs.yaml
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```
controlplane:~/dharani-k8s$ kubectl get pods -n dharani-ns
```

NAME	READY	STATUS	RESTARTS	AGE
ipl-rs-dzdcz	1/1	Running	0	95s
ipl-rs-hbwtb	1/1	Running	0	95s
ipl-rs-jg9gv	1/1	Running	0	95s
ipl-rs-pvxcn	1/1	Running	0	95s
ipl-rs-qtgt8	1/1	Running	0	95s

```
controlplane:~/dharani-k8s$ kubectl get rs -n dharani-ns
```

NAME	DESIRED	CURRENT	READY	AGE
ipl-rs	5	5	5	2m1s

```
controlplane:~/dharani-k8s$ kubectl get all -n dharani-ns
```

NAME	READY	STATUS	RESTARTS	AGE
pod/ipl-rs-dzdcz	1/1	Running	0	2m21s
pod/ipl-rs-hbwtb	1/1	Running	0	2m21s
pod/ipl-rs-jg9gv	1/1	Running	0	2m21s
pod/ipl-rs-pvxcn	1/1	Running	0	2m21s
pod/ipl-rs-qtgt8	1/1	Running	0	2m21s

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/ipl-rs	5	5	5	2m21s

```
controlplane:~/dharani-k8s$ kubectl delete pod ipl-rs-pvxcn -n dharani-ns
```

```
pod "ipl-rs-pvxcn" deleted from dharani-ns namespace
```

```
controlplane:~/dharani-k8s$ kubectl get all -n dharani-ns
```

NAME	READY	STATUS	RESTARTS	AGE
pod/ipl-rs-bwtm8	1/1	Running	0	29s
pod/ipl-rs-dzdcz	1/1	Running	0	3m45s
pod/ipl-rs-hbwtb	1/1	Running	0	3m45s

```
rs.yaml
```

```
controlplane:~/dharani-k8s$ kubectl get pods -n dharani-ns
```

NAME	READY	STATUS	RESTARTS	AGE
ipl-rs-dzdcz	1/1	Running	0	95s
ipl-rs-hbwtb	1/1	Running	0	95s
ipl-rs-jg9gv	1/1	Running	0	95s
ipl-rs-pvxcn	1/1	Running	0	95s
ipl-rs-qtgt8	1/1	Running	0	95s

```
controlplane:~/dharani-k8s$ kubectl get rs -n dharani-ns
```

NAME	DESIRED	CURRENT	READY	AGE
ipl-rs	5	5	5	2m1s

```
controlplane:~/dharani-k8s$ kubectl get all -n dharani-ns
```

NAME	READY	STATUS	RESTARTS	AGE
pod/ipl-rs-dzdcz	1/1	Running	0	2m21s
pod/ipl-rs-hbwtb	1/1	Running	0	2m21s
pod/ipl-rs-jg9gv	1/1	Running	0	2m21s
pod/ipl-rs-pvxcn	1/1	Running	0	2m21s
pod/ipl-rs-qtgt8	1/1	Running	0	2m21s

NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/ipl-rs	5	5	5	2m21s

```
controlplane:~/dharani-k8s$ kubectl delete pod ipl-rs-pvxcn -n dharani-ns
```

```
pod "ipl-rs-pvxcn" deleted from dharani-ns namespace
```

```
controlplane:~/dharani-k8s$ kubectl get all -n dharani-ns
```

NAME	READY	STATUS	RESTARTS	AGE
pod/ipl-rs-bwtm8	1/1	Running	0	29s
pod/ipl-rs-dzdcz	1/1	Running	0	3m45s
pod/ipl-rs-hbwtb	1/1	Running	0	3m45s

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controlplane:~/dharani-k8s$ kubectl describe rs ipl-rs -n dharani-ns
```

```
Name:          ipl-rs
Namespace:     dharani-ns
Selector:      app=ipl
Labels:        <none>
Annotations:   <none>
Replicas:      5 current / 5 desired
Pods Status:   5 Running / 0 Waiting / 0 Succeeded / 0 Failed
```

Pod Template:

```
Labels: app=ipl
Containers:
  ipl-container:
    Image:          muralisocial123/ipl:latest
    Port:           3000/TCP
    Host Port:      0/TCP
    Environment:    <none>
    Mounts:         <none>
    Volumes:        <none>
    Node-Selectors: <none>
    Tolerations:    <none>
```

Events:

Type	Reason	Age	From	Message
----	-----	----	----	-----
Normal	SuccessfulCreate	4m42s	replicaset-controller	Created pod: ipl-rs-pvxcn
Normal	SuccessfulCreate	4m42s	replicaset-controller	Created pod: ipl-rs-qtgt8
Normal	SuccessfulCreate	4m42s	replicaset-controller	Created pod: ipl-rs-jg9gv
Normal	SuccessfulCreate	4m42s	replicaset-controller	Created pod: ipl-rs-dzdcz
Normal	SuccessfulCreate	4m42s	replicaset-controller	Created pod: ipl-rs-hbwtb

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17 cat rs.yaml
18 kubectl apply -f rs.yaml
19 kubectl create ns -n dharani-ns
20 kubectl create ns dharani-ns
21 vi rs.yaml
22 kubectl apply -f rs.yaml
23 kubectl get rs -n dharani-ns
24 kubectl get pods -n dharani-ns
25 kubectl get all -n dharani-ns
26 ls
27 cd dharani-k8s/
28 ls
29 history
30 kubectl apply -f rs.yaml
31 vi rs.yaml
32 kubectl apply -f rs.yaml
33 kubectl delete rs ipl-rs -n dharani-ns
34 vi rs.yaml
35 kubectl apply -f rs.yaml
36 kubectl create ns -n dharani-ns
37 cat rs.yaml
38 la
39 kubectl get pods -n dharani-ns
40 kubectl get rs -n dharani-ns
41 kubectl get all -n dharani-ns
42 kubectl delete pod ipl-rs-pvxcn -n dharani-ns
43 kubectl describe rs ipl-rs -n dharani-ns
44 history
```

```

Selector:      app=ipl
Replicas:      5 desired | 5 updated | 5 total | 5 available | 0 unavailable
StrategyType:  RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=ipl
  Containers:
    ipl-container:
      Image:      muralisocial123/ipl:latest
      Port:       3000/TCP
      Host Port:  0/TCP
      Environment: <none>
      Mounts:      <none>
      Volumes:     <none>
      Node-Selectors: <none>
      Tolerations:  <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets: <none>
NewReplicaSet:  ipl-deployment-8445fb7bd6 (5/5 replicas created)
Events:
  Type    Reason             Age   From                  Message
  ----    -
  Normal  ScalingReplicaSet  4m16s deployment-controller Scaled up replica set ipl-deployment-8445fb7bd6 from 0 to 5

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Rollout:

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controlplane:~/dharani-k8s$ kubectl get all -n dharani-ns
NAME                                READY    STATUS    RESTARTS   AGE
pod/ipl-deployment-8445fb7bd6-54ftm 1/1      Running   0           6m34s
pod/ipl-deployment-8445fb7bd6-mhth5 1/1      Running   0           6m34s
pod/ipl-deployment-8445fb7bd6-mvwdv 1/1      Running   0           6m34s
pod/ipl-deployment-8445fb7bd6-wjx2s 1/1      Running   0           6m34s
pod/ipl-deployment-8445fb7bd6-wmlwg 1/1      Running   0           6m34s

NAME                                READY    UP-TO-DATE  AVAILABLE   AGE
deployment.apps/ipl-deployment      5/5      5           5           6m34s

NAME                                DESIRED  CURRENT  READY  AGE
replicaset.apps/ipl-deployment-8445fb7bd6 5        5        5      6m34s
controlplane:~/dharani-k8s$ kubectl describe pod ipl-deployment-8445fb7bd6-wmlwg -n murali-ns
Error from server (NotFound): namespaces "murali-ns" not found
controlplane:~/dharani-k8s$ kubectl describe pod ipl-deployment-8445fb7bd6-wmlwg -n dharani-ns
Name:          ipl-deployment-8445fb7bd6-wmlwg
Namespace:     dharani-ns
Priority:       0
Service Account: default
Node:          controlplane/172.30.1.2
Start Time:    Fri, 20 Feb 2026 02:47:47 +0000
Labels:        app=ipl
               pod-template-hash=8445fb7bd6
Annotations:   cni.projectcalico.org/containerID: 393696733d065155a3f1ce9606dfd35e0f64d055552bdce4ff92f37996236b6b
               cni.projectcalico.org/podIP: 192.168.0.4/32
               cni.projectcalico.org/podIPs: 192.168.0.4/32
Status:        Running

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Status:      Running
IP:          192.168.0.4
IPs:
  IP:        192.168.0.4
Controlled By: ReplicaSet/ipl-deployment-8445fb7bd6
Containers:
  ipl-container:
    Container ID:   containerd://40014d2de6c42206e17b457c3b1a142ef27d5cc39d50b5f5a4b56ccd6fe7ba98
    Image:          muralisocial123/ipl:latest
    Image ID:       docker.io/muralisocial123/ipl@sha256:77030f01a2375683dde323945e1752d7d7d145e6cbd4b431f4c1b4b32d811
7a5
    Port:          3000/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Fri, 20 Feb 2026 02:48:37 +0000
    Ready:         True
    Restart Count:  0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-lpttx (ro)
Conditions:
  Type                                Status
  PodReadyToStartContainers          True
  Initialized                         True
  Ready                              True
  ContainersReady                    True
  PodScheduled                       True
Volumes:
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Status:      Running
IP:          192.168.0.4
IPs:
  IP:        192.168.0.4
Controlled By: ReplicaSet/ipl-deployment-8445fb7bd6
Containers:
  ipl-container:
    Container ID:   containerd://40014d2de6c42206e17b457c3b1a142ef27d5cc39d50b5f5a4b56ccd6fe7ba98
    Image:          muralisocial123/ipl:latest
    Image ID:       docker.io/muralisocial123/ipl@sha256:77030f01a2375683dde323945e1752d7d7d145e6cbd4b431f4c1b4b32d811
7a5
    Port:          3000/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Fri, 20 Feb 2026 02:48:37 +0000
    Ready:         True
    Restart Count:  0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-lpttx (ro)
Conditions:
  Type                                Status
  PodReadyToStartContainers          True
  Initialized                         True
  Ready                              True
  ContainersReady                    True
  PodScheduled                       True
Volumes:
```

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```

Volumes:
  kube-api-access-lpttx:
    Type: Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName: kube-root-ca.crt
    Optional: false
    DownwardAPI: true
QoS Class: BestEffort
Node-Selectors: <none>
Tolerations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
              node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type      Reason      Age   From          Message
  ----      -
Normal     Scheduled   7m47s default-scheduler Successfully assigned dharani-ns/ipl-deployment-8445fb7bd6-wmlwg to controlplane
Normal     Pulling     7m45s kubelet        Pulling image "muralisocial123/ipl:latest"
Normal     Pulled      6m57s kubelet        Successfully pulled image "muralisocial123/ipl:latest" in 47.993s (47.94s including waiting). Image size: 445562584 bytes.
Normal     Created     6m57s kubelet        Created container: ipl-container
Normal     Started     6m57s kubelet        Started container ipl-container
controlplane:~/dharani-k8s$ kubectl get all -n dharani-ns
NAME                                READY    STATUS    RESTARTS   AGE
pod/ipl-deployment-8445fb7bd6-54ftm  1/1      Running   0           8m33s
pod/ipl-deployment-8445fb7bd6-mhth5  1/1      Running   0           8m33s
pod/ipl-deployment-8445fb7bd6-mvwvdv 1/1      Running   0           8m33s
pod/ipl-deployment-8445fb7bd6-wjx2s  1/1      Running   0           8m33s
pod/ipl-deployment-8445fb7bd6-wmlwg   1/1      Running   0           8m33s

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```

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/ipl-deployment      5/5      5              5            8m33s

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/ipl-deployment-8445fb7bd6  5          5          5        8m33s
controlplane:~/dharani-k8s$ kubectl rollout undo ipl-deployment -n dharani-ns
error: the server doesn't have a resource type "ipl-deployment"
controlplane:~/dharani-k8s$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ipl-deployment
  namespace: dharani-ns
spec:
  replicas: 5
  selector:
    matchLabels:
      app: ipl
  template:
    metadata:
      labels:
        app: ipl
    spec:
      containers:
        - name: ipl-container
          image: muralisocial123/ipl:latest
          ports:

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    ports:
      - containerPort: 3000
controlplane:~/dharani-k8s$ kubectl rollout history deployment ipl-deployment -n dharani-ns
deployment.apps/ipl-deployment
REVISION  CHANGE-CAUSE
1          <none>

controlplane:~/dharani-k8s$ kubectl rollout status deployment ipl-deployment -n dharani-ns
deployment "ipl-deployment" successfully rolled out
controlplane:~/dharani-k8s$ kubectl get all -n dharani-ns
NAME                                READY   STATUS    RESTARTS   AGE
pod/ipl-deployment-8445fb7bd6-54ftm 1/1     Running   0           11m
pod/ipl-deployment-8445fb7bd6-mhth5 1/1     Running   0           11m
pod/ipl-deployment-8445fb7bd6-mvw dv 1/1     Running   0           11m
pod/ipl-deployment-8445fb7bd6-wjx2s 1/1     Running   0           11m
pod/ipl-deployment-8445fb7bd6-wmlwg 1/1     Running   0           11m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/ipl-deployment      5/5     5             5           11m

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/ipl-deployment-8445fb7bd6 5         5         5       11m
controlplane:~/dharani-k8s$ kubectl describe deployment ipl-deployment -n dharani-ns
Name:          ipl-deployment
Namespace:     dharani-ns
CreationTimestamp: Fri, 20 Feb 2026 02:47:47 +0000
Labels:        <none>
Annotations:    deployment.kubernetes.io/revision: 1

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controlplane:~/dharani-k8s$ kubectl describe pod ipl-deployment-8445fb7bd6-wjx2s -n dharani-ns
Name:          ipl-deployment-8445fb7bd6-wjx2s
Namespace:     dharani-ns
Priority:       0
Service Account: default
Node:          node01/172.30.2.2
Start Time:    Fri, 20 Feb 2026 02:47:47 +0000
Labels:        app=ipl
               pod-template-hash=8445fb7bd6
Annotations:    cni.projectcalico.org/containerID: c06622b92759abd3cff9fd159667c3ad007051488a84212d598f7bf793572a7d
               cni.projectcalico.org/podIP: 192.168.1.4/32
               cni.projectcalico.org/podIPs: 192.168.1.4/32
Status:        Running
IP:            192.168.1.4
IPs:           IP: 192.168.1.4
Controlled By: ReplicaSet/ipl-deployment-8445fb7bd6
Containers:
  ipl-container:
    Container ID: containerd://6e52d1da1ac86c2a2a19f22ccc9ab3a29ac106cd95aaa7a06a955d480364cd02
    Image:        muralisocial123/ipl:latest
    Image ID:     docker.io/muralisocial123/ipl@sha256:77030f01a2375683dde323945e1752d7d7d145e6cbd4b431f4c1b4b32d811
    Port:        3000/TCP
    Host Port:    0/TCP
    State:        Running
      Started:    Fri, 20 Feb 2026 02:48:21 +0000
    Ready:        True

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controlplane:~/dharani-k8s$ kubectl describe pod ipl-deployment-8445fb7bd6-wjx2s -n dharani-ns
Name:          ipl-deployment-8445fb7bd6-wjx2s
Namespace:     dharani-ns
Priority:       0
Service Account: default
Node:          node01/172.30.2.2
Start Time:    Fri, 20 Feb 2026 02:47:47 +0000
Labels:        app=ipl
                pod-template-hash=8445fb7bd6
Annotations:   cni.projectcalico.org/containerID: c06622b92759abd3cff9fd159667c3ad007051488a84212d598f7bf793572a7d
                cni.projectcalico.org/podIP: 192.168.1.4/32
                cni.projectcalico.org/podIPs: 192.168.1.4/32
Status:        Running
IP:            192.168.1.4
IPs:
  IP:          192.168.1.4
Controlled By: ReplicaSet/ipl-deployment-8445fb7bd6
Containers:
  ipl-container:
    Container ID: containerd://6e52d1da1ac86c2a2a19f22ccc9ab3a29ac106cd95aaa7a06a955d480364cd02
    Image:        muralisocial123/ipl:latest
    Image ID:     docker.io/muralisocial123/ipl@sha256:77030f01a2375683dde323945e1752d7d7d145e6cbd4b431f4c1b4b32d811
    Port:         3000/TCP
    Host Port:    0/TCP
    State:        Running
      Started:    Fri, 20 Feb 2026 02:48:21 +0000
    Ready:        True

```

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Type                                         Status
PodReadyToStartContainers                  True
Initialized                               True
Ready                                      True
ContainersReady                            True
PodScheduled                              True
Volumes:
  kube-api-access-bqxjz:
    Type:          Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:  kube-root-ca.crt
    Optional:       false
    DownwardAPI:    true
QoS Class:           BestEffort
Node-Selectors:      <none>
Tolerations:         node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                     node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason      Age   From          Message
  ----     -
Normal    Scheduled   14m   default-scheduler   Successfully assigned dharani-ns/ipl-deployment-8445fb7bd6-wjx2s to node
01
Normal    Pulling     14m   kubelet        Pulling image "muralisocial123/ipl:latest"
Normal    Pulled      13m   kubelet        Successfully pulled image "muralisocial123/ipl:latest" in 33.042s (33.04
2s including waiting). Image size: 445562584 bytes.
Normal    Created     13m   kubelet        Created container: ipl-container
Normal    Started     13m   kubelet        Started container ipl-container
controlplane:~/dharani-k8s$

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19 history
20 kubectl get all -n dharani-ns
21 kubectl describe pod ipl-deployment-8445fb7bd6-wmlwg -n murali-ns
22 kubectl describe pod ipl-deployment-8445fb7bd6-wmlwg -n dharani-ns
23 kubectl get all -n dharani-ns
24 kubectl rollout undo ipl-deployment -n dharani-ns
25 cat deployment.yaml
26 kubectl rollout history deployment ipl-deployment -n dharani-ns
27 kubectl rollout status deployment ipl-deployment -n dharani-ns
28 kubectl get all -n dharani-ns
29 kubectl describe pod pod/ipl-deployment-8445fb7bd6-wjx2s -n dharani-ns
30 kubectl describe pod ipl-deployment-8445fb7bd6-wjx2s -n dharani-ns
31 history
controlplane:~/dharani-k8s$
```